

Practical 8

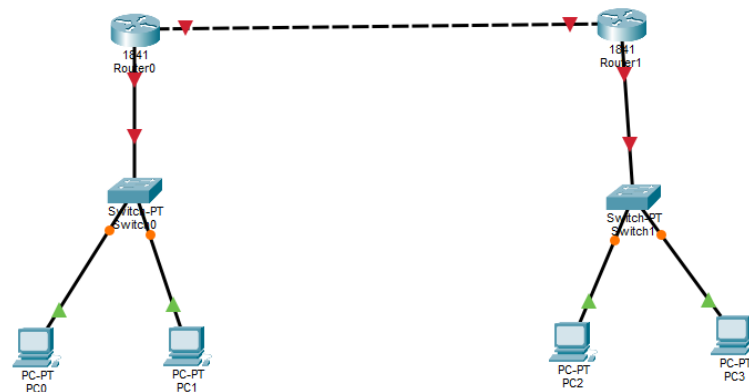
Static routing: Is a technique where a network administrator manually configure fixed path in router's routing table.

Steps to implement static routing :

- 1) Add the PCs and switches and make two local area network and then add two router for both the local area network.



- 2) Connect all PCs , switches and Router by copper cable.



- 3) Provide IP address to all the Pcs and to both the routers (to both Fat Ethernet cable) and also on the port status for both the cables.

PC4

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:43FF:FE65:9172

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

Router0

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

FastEthernet0/0

Port Status ☒ On

Link Speed 100 Mbps 10 Mbps ☒ Auto

Duplex Half Duplex Full Duplex ☒ Auto

MAC Address 000D.BD02.CA01

IP Configuration

IPv4 Address 192.168.10.1

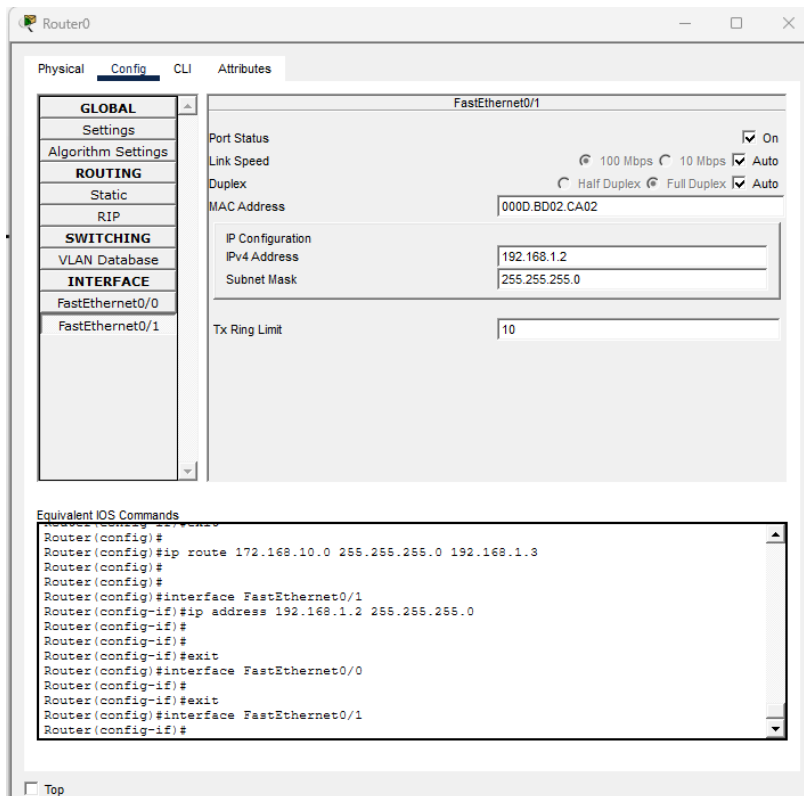
Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

```
*LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#
Router(config)#ip route 172.168.10.0 255.255.255.0 192.168.1.3
Router(config)#
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.1.2 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
```

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- 4) Go to configuration and go to routing -> static in network add the ip address of local area network of another network and mask as 255.255.255.0 and add if address of another router in next hop. Same configuration for another router .

Router0

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

Static Routes

Network 172.168.10.0

Mask 255.255.255.0

Next Hop 192.168.1.3

Add

Network Address

172.168.10.0/24 via 192.168.1.3

Remove

Equivalent IOS Commands

```

Router(config)#
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.1.2 255.255.255.0
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#

```

Router1

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

Static Routes

Network 192.168.10.0

Mask 255.255.255.0

Next Hop 192.168.1.2

Add

Network Address

192.168.10.0/24 via 192.168.1.2

Remove

Equivalent IOS Commands

```

Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ip address 192.168.1.3 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#ip route 192.168.10.0 255.255.255.0 192.168.1.2
Router(config)#
Router(config)#
Router(config)#
Router(config)#

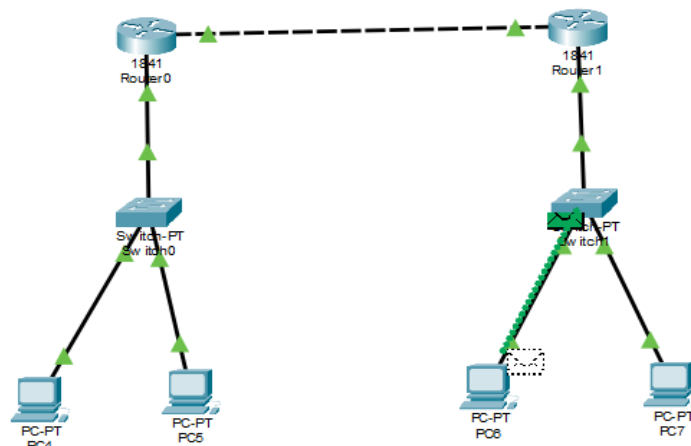
```

- 5) Go to the lens and see the router table by moving lens to router.

Type	Network	Port	Next Hop IP	Metric
S	172.168.10.0/24	---	192.168.1.3	1/0
C	192.168.1.0/24	FastEthernet0/1	---	0/0
C	192.168.10.0/24	FastEthernet0/0	---	0/0

Type	Network	Port	Next Hop IP	Metric
C	172.168.10.0/24	FastEthernet0/0	---	0/0
C	192.168.1.0/24	FastEthernet0/1	---	0/0
S	192.168.10.0/24	---	192.168.1.2	1/0

- 6) Check the connection by sending the packet from local area of one router to another



b) **RIP routing:** A networking technique where routers automatically learn and update network paths using protocols.

Steps to implement RIP routing :

- 1) Make the connection as above then go to routing->RIP then add networks for both the routers.

Router0

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

RIP Routing

Network

Network Address

192.168.1.0

192.168.10.0

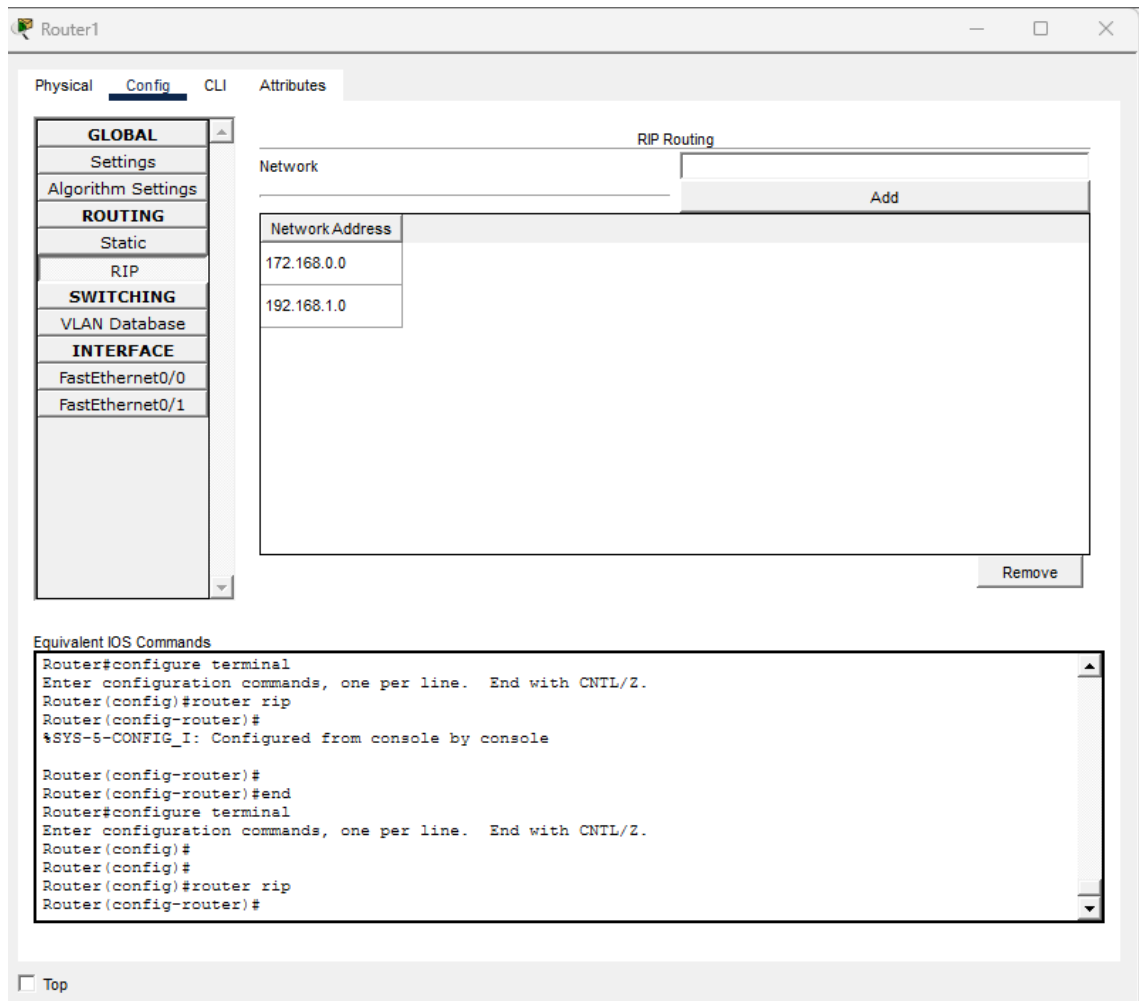
Add

Remove

Equivalent IOS Commands

Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.1.2 255.255.255.0
Router(config-if)#
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#no network 192.168.10.0
Router(config-router)#network 192.168.10.0
Router(config-router)#

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2) Check the connection by sending the packet from local area of one router to another

